HDOC DAY-52

PRODUCT EXCEPT SELF USING POINTERS

Aim

To write a C program that finds the product of all array elements except the current element, without using division, and passes the array to a function using pointers.

Problem Statement

- Read the size and elements of a variable-length integer array, nums[].
- Pass the array to product_except_self() using its address and pointers.
- Create an output array answer[] of the same size as nums[].
- Store in answer[i] the product of all elements except nums[i].
- Do not use division. Use prefix and suffix products instead.
- Display the result in the original order, separated by commas.

Algorithm

1. Read n and the n elements of nums[].

2. Initialize answer[0] = 1.

3. Move from left to right and store the product of all elements before each position in answer[].

4. Initialize suffix = 1.

5. Move from right to left. Multiply answer[i] by suffix, then update suffix using nums[i].

6. Print answer[] separated by commas.

C Program

```c
#include <stdio.h>

void product_except_self(int *nums, int n)
{
    int answer[n];
    int suffix = 1;
    int i;

    answer[0] = 1;

    for (i = 1; i < n; i++)
        answer[i] = answer[i - 1] * *(nums + i - 1);

    for (i = n - 1; i >= 0; i--)
    {
        answer[i] = answer[i] * suffix;
        suffix = suffix * *(nums + i);
    }

    printf("Output: ");
    for (i = 0; i < n; i++)
    {
        printf("%d", answer[i]);
        if (i < n - 1)
            printf(", ");
    }
    printf("\n");
}

int main()
{
    int n, i;

    printf("Enter size of array: ");
    scanf("%d", &n);

    int nums[n];

    printf("Enter array elements: ");
    for (i = 0; i < n; i++)
        scanf("%d", nums + i);

    product_except_self(nums, n);

    return 0;
}
```

Test Cases

Test Case	Input Array	Expected Output
1	1 2 3 4	24, 12, 8, 6
2	-1 1 0 -3 3	0, 0, 9, 0, 0

Sample Output - Test Case 1

```
Enter size of array: 4
Enter array elements: 1 2 3 4
Output: 24, 12, 8, 6
```

Sample Output - Test Case 2

```
Enter size of array: 5
Enter array elements: -1 1 0 -3 3
Output: 0, 0, 9, 0, 0
```

Result

The program successfully calculates the product of all elements except the current element without using division. The array is passed to the function using pointers, and the result is displayed in the required comma-separated format.